BASIC MATH

Pertemuan 1

Study Group SCC

KITA BELAJAR APA SIH??

Belajar dasarnya dulu ya sebelum jump in ke advanced part

01 **Aljabar Linear**
Eigenvalue & Eigenvector

02 **Kalkulus**
Turunan & Turunan Parsial

03 **Statistika**
Linear Regression & MLE

ALJABAR LINEAR

Eigenvalue & Eigenvector

Eigen vector dan eigen value berperan di balik layar dalam banyak teknik reduksi dimensi, seperti analisis komponen utama (PCA). Selain PCA, eigen value juga umum digunakan untuk *signal processing* dan *data transmission*

Vektor-vektor yang tidak berubah arah (atau berputar ke arah yang berlawanan) adalah vektor eigen, karena mereka menjadi vektor "karakteristik" dari transformasi yang terjadi. Nilai eigen adalah faktor perbesaran dari transformasi tersebut.

$$A\mathbf{v} = \lambda\mathbf{v}$$

A = matrix *n* x *n*

v = eigen vector

λ = eigen value

Tipe *eigenvector*

- *right* → gunakan **v**
- *left* → gunakan $\mathbf{v}^{\mathbf{T}}$

Umumnya menggunakan *right eigenvector.*

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ . \\ . \\ v_n \end{pmatrix}$$

$$\mathbf{v}^{\mathbf{T}} = (v_1 \quad v_2 \quad . \quad . \quad v_n)$$

$$I\lambda = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

Rumus Dasar
HARUS DIPAHAMI!!

$$A\vec{v} = (I\lambda)\vec{v}$$

$$A\vec{v} - I\lambda\vec{v} = 0$$

$$(A - I\lambda)\vec{v} = 0$$

BAGAIMANA CARANYA?

Menghitung eigenvalue & eigenvektor

1. Persamaan transformasi

$$A \cdot v - \lambda \cdot v = 0$$

2. *Rearrange* (ubah bentuk)

$$(A - \lambda \cdot I) \cdot v = 0$$

3. Cari determinan

$$\det(A - \lambda \cdot I) = 0$$

4. Cari *eigenvalue*

Misal A matrix 2x2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

$$\det\left(\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}\right) = 0$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

5. Cari *eigenvector*

Masukkan nilai *eigenvalue* yang didapat dari no (4) ke dalam persamaan berikut :

$$(A - \lambda \cdot I) \cdot v = 0$$

KALKULUS

Dasar Turunan

01 $f(x) = c$
$f'(x) = 0.$

c merupakan konstanta

02 $h(x)=f(g(x))$
$h'(x)=f'(g(x))\cdot g'(x)$

Ini aturan rantai

03 $f(x) = k.x$
$f'(x) = k$

04 $h(x) = u(x).\ v(x)$
$h'(x) = u'(x).\ v(x) + u(x).\ v'(x)$

05 $h(x) = u(x)\ /\ v(x)$
$h'(x) = (\ u'v - uv'\ )\ /\ v^2$

06 $f(x) = ax^n$
$f'(x) = anx^{n-1}$

KALKULUS

Contoh Aturan Rantai

$y = (3x + 2)^5$

Misalkan $u = 3x + 2$

$y = u^5$

$\frac{dy}{du} = 5u^4 \qquad \frac{du}{dx} = 3$

$\frac{dy}{dx} = 5u^4 \cdot 3$

$= 15(3x + 2)^4$

$y = \sin(2x^2)$

Misalkan $u = 2x^2$

$\frac{dy}{du} = \cos(\mathrm{u}) \qquad \frac{du}{dx} = 4\mathrm{x}$

$\frac{dy}{dx} = \cos(2x^2) \cdot 4x$

$= 4x\cos(2x^2)$

$y = e^{x^2+3x}$

Misalkan $u = x^2 + 3x$

$y = e^u$

$\frac{dy}{du} = e^u \qquad \frac{du}{dx} = 2x + 3$

$\frac{dy}{dx} = e^{x^2+3x}(2x + 3)$

KALKULUS

Turunan Implisit

Fungsi **tidak dalam bentuk eksplisit** y=f(x)

Turunkan 2 sisi terhadap x

INGAT BENTUK DI SAMPING :

$$\frac{d}{dx}(y) = \frac{dy}{dx}$$

Contoh :

$$x^2 + y^2 = 25$$

$$2x + 2y\frac{dy}{dx} = 0$$

$$2y\frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

KALKULUS

Turunan Parsial

Digunakan ketika fungsinya **lebih dari satu variabel.**

Misal 2 variabel : $f(x,y)$

Turunkan terhadap tiap variabel

Turunkan terhadap x $\rightarrow$ y konstan

Turunkan terhadap y $\rightarrow$ x konstan

Contoh :

$$f(x,y) = x^2y + 3xy$$

Fungsi 2 variabel

$$\frac{\partial f}{\partial x} = 2xy + 3y$$

Turunan terhadap x

$$\frac{\partial f}{\partial y} = x^2 + 3x$$

Turunan terhadap y

Turunan Fungsi Eksponen dan Logaritma Natural

1. $y = a^x \rightarrow y' = a^x . \ln a$
2. $y = a^{u(x)} \rightarrow y' = a^{u(x)} . \ln a . u'(x)$
3. $y = e^x \rightarrow y' = e^x$
4. $y = e^{u(x)} \rightarrow y' = e^{u(x)} . u'(x)$
5. $y = \log_a x \rightarrow y' = \frac{1}{x} . \frac{1}{\ln a}$
6. $y = \log_a u(x) \rightarrow y' = \frac{1}{u(x)} . \frac{1}{\ln a} . u'(x)$
7. $y = \ln x \rightarrow y' = \frac{1}{x}$
8. $y = \ln u(x) \rightarrow y' = \frac{1}{u(x)} . u'(x)$

Turunan Fungsi Trigonometri

1. $f(x) = \sin x \Leftrightarrow f'(x) = \cos x$
2. $f(x) = \cos x \Leftrightarrow f'(x) = -\sin x$
3. $f(x) = \tan x \Leftrightarrow f'(x) = \sec^2 x$
4. $f(x) = \cot x \Leftrightarrow f'(x) = -\csc^2 x$
5. $f(x) = \sec x \Leftrightarrow f'(x) = \sec x \tan x$
6. $f(x) = \csc x \Leftrightarrow f'(x) = -\csc x \cot x$

STATISTIKA

Regresi

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Rumus dasar regresi linear

$$b_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$b_0 = \frac{\sum y_i - b_1 \sum x_i}{n}$$

Sama saja dengan rumus dasar, hanya saja dipecah

$$y = \beta_0 + \beta_1 x + \varepsilon$$

$$\hat{y} = b_0 + b_1 x$$

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

Bentuk matriks

STATISTIKA

MLE pada regresi linear

MLE adalah metode yang mencari nilai parameter yang **memaksimalkan fungsi likelihood**, yaitu fungsi yang menggambarkan seberapa besar kemungkinan data yang diamati dapat muncul berdasarkan nilai parameter tertentu.

MLE dapat digunakan untuk mengestimasi parameter regresi linear. Kali ini kita akan menggunakan model matrix

Bentuk dasar persamaan regresi linear

$$\mathbf{Y}_{n\times 1} = \mathbf{X}_{n\times(p+1)}\boldsymbol{\beta}_{(p+1)\times 1} + \boldsymbol{\varepsilon}_{n\times 1}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1p} \\ 1 & X_{21} & X_{22} & \cdots & X_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

$$= \begin{bmatrix} \beta_0 + \beta_1 X_{11} + \beta_2 X_{12} + \cdots + \beta_p X_{1p} + \varepsilon_1 \\ \beta_0 + \beta_1 X_{21} + \beta_2 X_{22} + \cdots + \beta_p X_{2p} + \varepsilon_2 \\ \vdots \\ \beta_0 + \beta_1 X_{n1} + \beta_2 X_{n2} + \cdots + \beta_p X_{np} + \varepsilon_n \end{bmatrix}$$

Rumus dasar distribusi normal

Variabel acak y_i dengan distribusi data sampel normal

$$y_i \sim N(\mu, \sigma^2)$$

$$f(y_i) = (2\pi\sigma^2)^{-1/2} \exp\left[-\frac{(y_i-\mu)^2}{2\sigma^2}\right]$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i-\mu)^2}{2\sigma^2}\right]$$

Asumsi normalitas, (εi) berdistribusi normal dengan rata-rata (0) dan ragam (σ^2), sehingga :

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$f(\varepsilon_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(\varepsilon_i-0)^2}{2\sigma^2}\right]$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\varepsilon_i^2}{2\sigma^2}\right]$$

PDF pada regresi linear berganda

Ingat Y = Xβ + ε , maka ε = Y - Xβ. Masukkan ke persamaan pada slide sebelumnya

$$f(y_i \mid x_i, \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{p}(y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_p X_{ip})^2\right]$$

$$f(y \mid x, \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right]$$

$$= (2\pi\sigma^2)^{-1/2} \exp\left[-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right]$$

Likelihood function, Log - Likelihood, & Estimasi Parameter

$$L(y_i \mid x_i, \beta, \sigma^2) = \prod_{i=1}^{n} f(y_i \mid x_i, \beta, \sigma^2)$$

$$L(y \mid x, \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right]$$

$$\ln L(y \mid x, \beta, \sigma^2) = \ln\left[(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right)\right]$$

$$= \ln(2\pi\sigma^2)^{-n/2} + \ln\left[\exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right)\right]$$

$$= -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$\frac{\partial \ln L(y \mid x, \beta, \sigma^2)}{\partial \boldsymbol{\beta}} = 0$$

$$-\frac{1}{2\sigma^2}(-2\mathbf{X}'\mathbf{Y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta}) = 0$$

$$-2\mathbf{X}'\mathbf{Y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = 0$$

$$-\mathbf{X}'\mathbf{Y} + \mathbf{X}'\mathbf{X}\boldsymbol{\beta} = 0$$

$$\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{Y}$$

$$\boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Thank you

References

- http://repository.uki.ac.id/6168/1/Turunan.pdf
- https://www.geeksforgeeks.org/engineering-mathematics/eigen-values/
- https://www.datacamp.com/tutorial/eigenvectors-eigenvalues
- https://www.onlinemathlearning.com/chain-rule.html
- https://repository.ub.ac.id/id/eprint/168490/1/DIAH%20AYU%20WIDYASTUTI.pdf